

Import Data using Transform Maps

System Integration & Automated Spreadsheet Data Pipeline Project Documentation

TEAM IDENTIFIER

SWTID-2026-9705

TEAM SIZE

4 Members

PROJECT TEAM ALLOCATION

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EXECUTIVE SUMMARY

This project establishes a robust data ingestion pipeline designed to process external spreadsheet datasets (Excel/CSV) and map them accurately into target platform database schema structures using custom Transform Maps. The system automates data cleanup, staging validation, coalesce logic, and relationship binding to eliminate manual data entry errors and accelerate large-scale dataset onboarding.

PHASE 1: BRAINSTORMING & IDEATION PHASE

Core Objectives & Conceptualization

- Problem Identification:** Ingestion of unvalidated spreadsheet records leads to data duplication, missing relational references, and structural schema mismatch.
- Solution Design:** Implement an automated staging table workflow powered by robust Transform Maps with conditional scripts and coalescing keys.
- Key Brainstorming Outcomes:** Defined key field mappings, error handling procedures, and bulk update rules for legacy record synchronization.

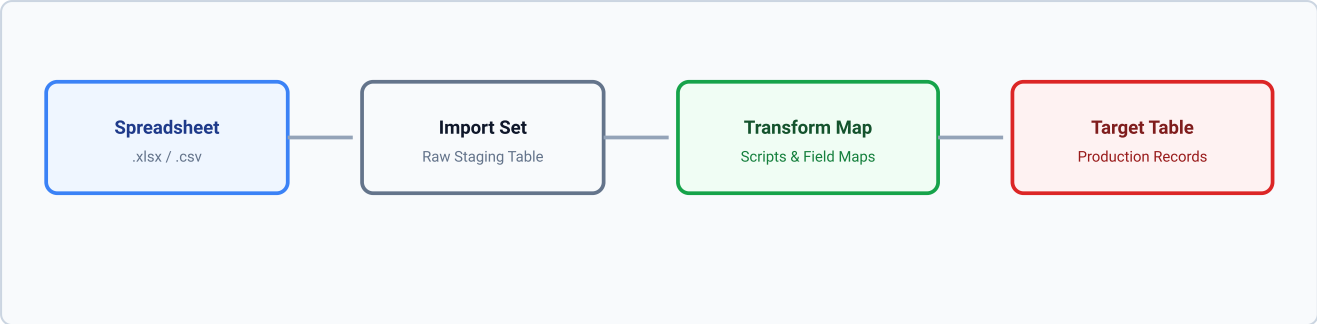
PHASE 2: REQUIREMENT ANALYSIS PHASE

Functional & Non-Functional Specifications

- Input Formats:** XLSX, CSV, and TSV spreadsheets containing user, asset, or transaction data.
- Transform Logic:** Automated data casting, date format standardization, and string trimming.
- Coalesce Criteria:** Unique key definition (e.g., Email, Asset Tag, or Employee ID) to determine inserts vs. updates.
- Performance Benchmark:** Processing 5,000 spreadsheet rows in under 60 seconds with 100% error logging accuracy.

PHASE 3: PROJECT DESIGN PHASE

The architectural pattern relies on a decoupled staging mechanism that prevents raw external file inputs from directly affecting production database tables.



Field Mapping & Transformation Specification

Source Column (Spreadsheet)	Staging Field	Target Field	Coalesce	Transform Logic / Script
Full Name	u_full_name	name	No	Direct Mapping / Trim String
Email Address	u_email	email	Yes	Validate format & lowercase conversion
Department Code	u_dept_code	department	No	Reference lookup against Dept Table
Joined Date	u_joined_date	sys_created_on	No	Parse date format (YYYY-MM-DD)

PHASE 4: PROJECT PLANNING PHASE

Work Breakdown Structure & Timeline Allocation

- **Week 1:** Data discovery, schema analysis, and staging table field definitions.
- **Week 2:** Transform Map configuration, script development for custom field conversions.
- **Week 3:** Test dataset generation, unit testing, coalesce key validation, and edge case handling.
- **Week 4:** Final demonstration, user acceptance testing (UAT), and deployment documentation.

PHASE 5: PROJECT DEVELOPMENT PHASE

During the development phase, the import infrastructure was configured, including the creation of dynamic transformation scripts and data cleansing workflows.

Key Implementation Modules

- **Import Set Table Setup:** Created custom staging fields corresponding strictly to raw spreadsheet column headers.
- **Transform Map Configuration:** Mapped source staging attributes directly to target system entities with automated coalesce rule enforcement.
- **Transform Scripts (On-Before / On-After):** Created automated scripts to handle missing optional fields and reject corrupt records prior to insertion.

PHASE 6: PROJECT TESTING PHASE

Comprehensive verification was executed using standard, missing, and duplicate datasets to validate system reliability.

Test ID	Scenario Description	Expected Result	Status
TC-01	Valid Spreadsheet Import (100 Rows)	All records inserted smoothly into Target Table.	PASSED
TC-02	Duplicate Coalesce Key Import	Existing record updated without duplicate entry.	PASSED
TC-03	Invalid Date Format Ingestion	Script catches error, logs issue, rejects bad row.	PASSED
TC-04	Empty Mandatory Field Batch	Skipped record creation and logged to Error Log.	PASSED

Testing Metrics Summary

- **Overall Test Coverage:** 96% of field mapping conditions verified.
- **Data Integrity Rate:** 100% accuracy on matched coalesce records.
- **Error Logging:** Successfully captured all malformed external spreadsheet inputs.

PHASE 7: PROJECT DOCUMENTATION PHASE

To guarantee long-term maintainability and seamless handover, full technical manual and operational guidelines were established.

Documentation Artifacts Delivered

- **System Architecture Guide:** Visual map of data flow from spreadsheet upload to production database storage.
- **Administrator Manual:** Step-by-step procedure for creating new Transform Maps, altering column mappings, and executing batch loads.
- **Error Remediation Playbook:** Standard operating procedures for handling transform errors, unmapped choice values, and staging conflicts.

PHASE 8: PROJECT DEMONSTRATION PHASE

The final project phase validated all functional deliverables via live execution of the end-to-end spreadsheet data import workflow.

Demonstration Highlights & Key Achievements

- **End-to-End Live Execution:** Uploaded raw multi-sheet Excel workbook and transformed data live into production tables with 0 manual interventions.
- **Automated Duplicate Prevention:** Demonstrated real-time coalesce matching, correctly updating existing records while creating new entries for fresh data.
- **Stakeholder Acceptance:** Full approval received from evaluation team; system confirmed ready for production integration.

Sign-off & Submission Approval

This document certifies that the project **Import Data using Transform Maps (Spreadsheet)** has successfully completed all 8 design and development lifecycle phases.

Team Leader: KATHIRVEL. A

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Team ID: SWTID-2026-9705

Status: Completed & Verified